Workshop on Machine Learning for Earth System Modelling
24–26 August 2026 · Bonn, Germany

Latent Windhose:

Mechanistic Analysis of GraphCast Latent Space

Linear Analysis of the Internal Activations
in a Data-Driven Weather Model

MALIN BRAATZ, SABINE SCHOLLE, ROBIN RAIWEL, PASCAL NIETERS | OSNABRÜCK UNIVERSITY

The Big Question:

Are data-driven ML weather models learning interpretable physical features?

MALIN BRAATZ, SABINE SCHOLLE | OSNABRÜCK UNIVERSITY

1

Mapping Latent Space Directions to Physical Space

2

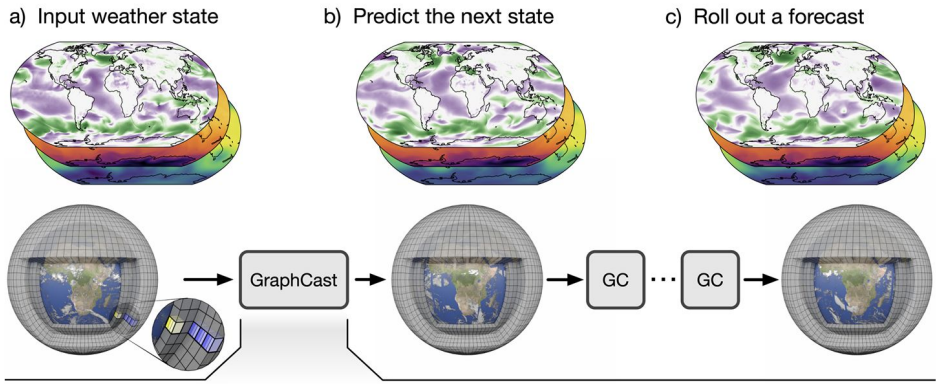
Finding Weather Events in Latent Space

3

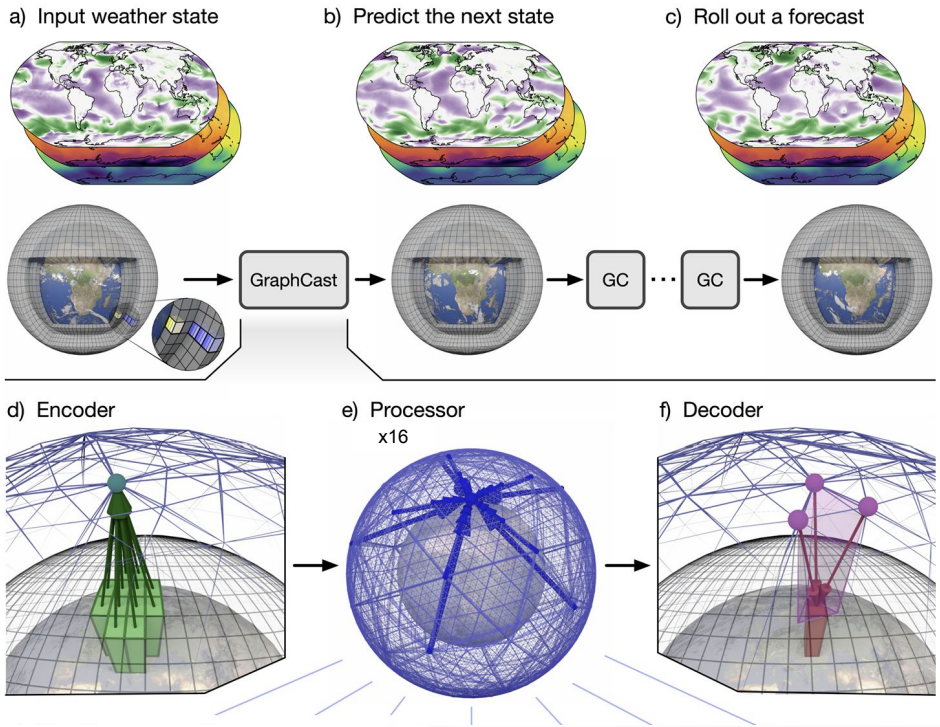
Perturbation of Weather Events in Latent Space

Mapping

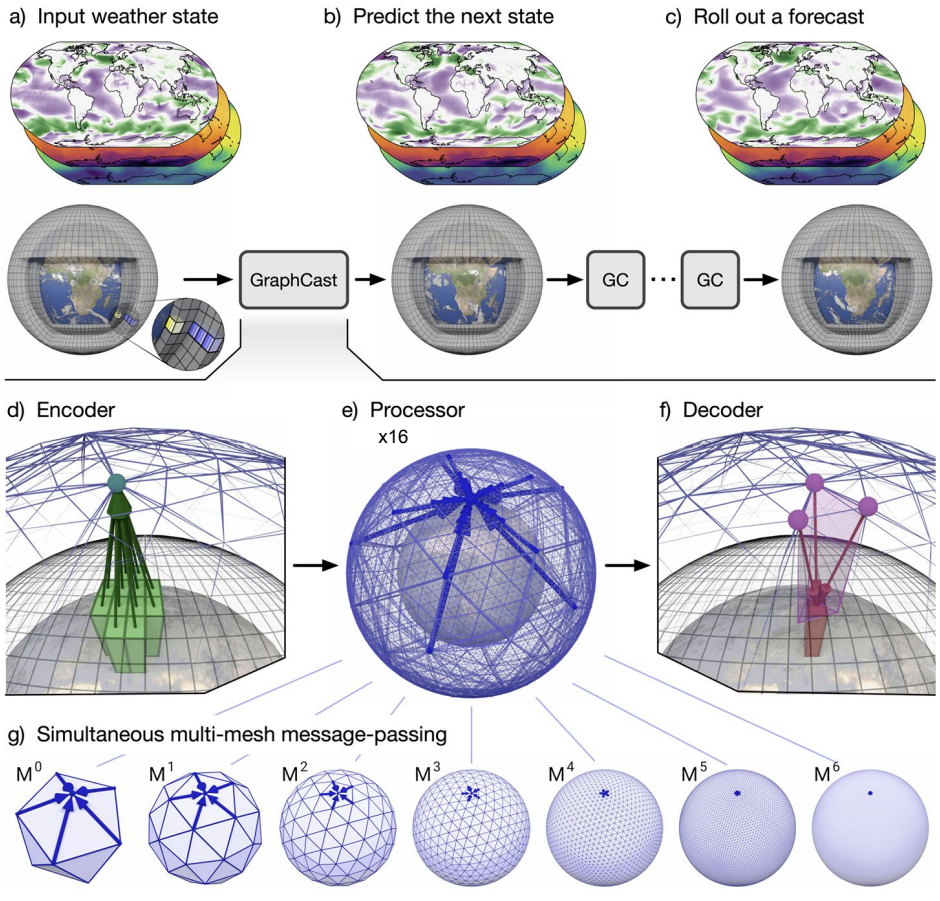
MALIN BRAATZ, SABINE SCHOLLE | OSNABRÜCK UNIVERSITY



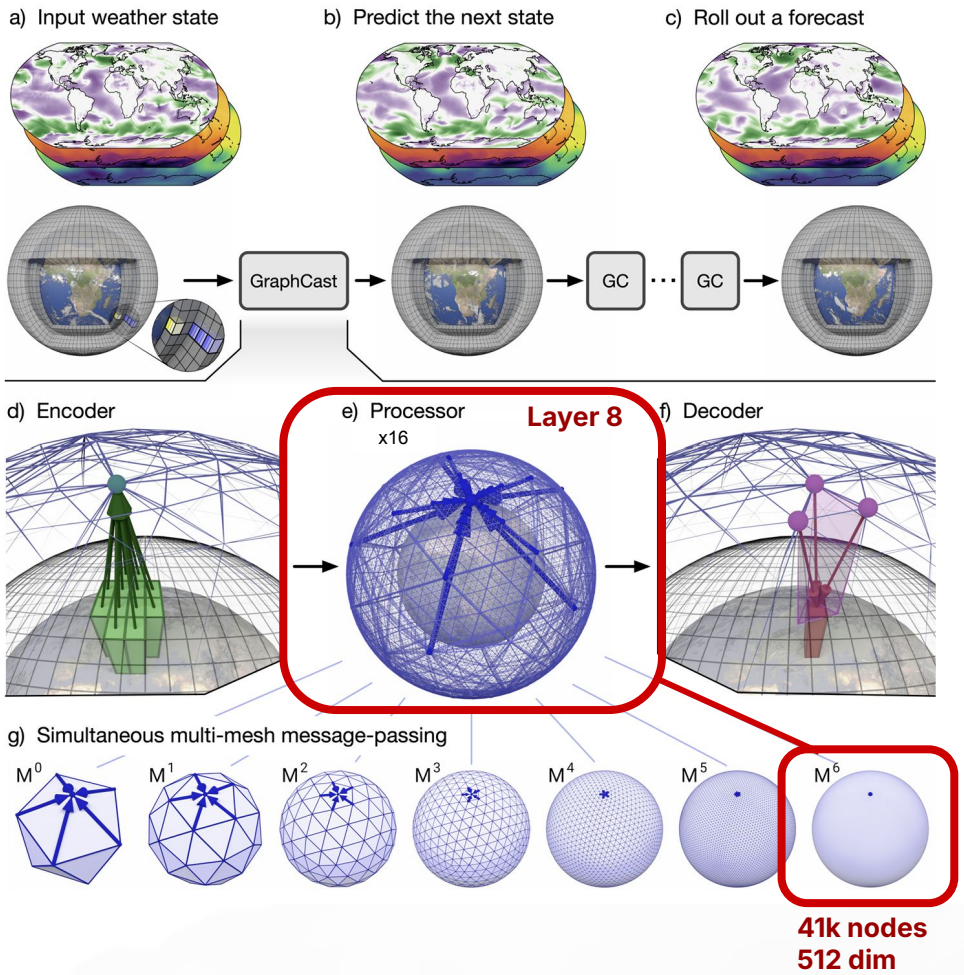
Lam et al. (2022)



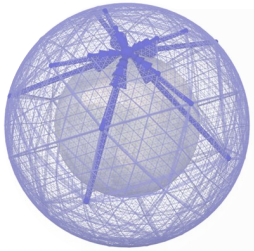
Lam et al. (2022)



Lam et al. (2022)

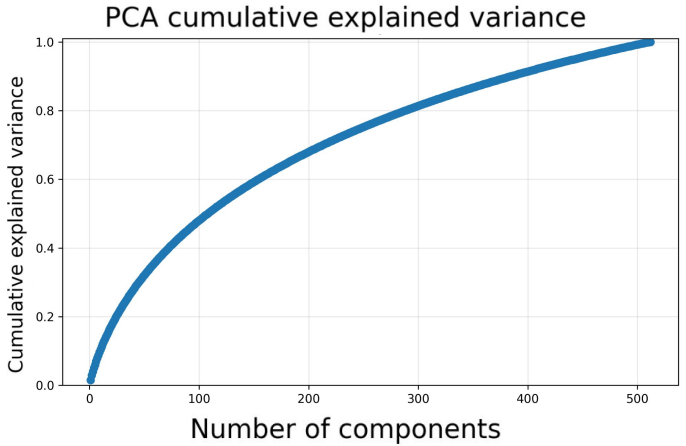


Lam et al. (2022)

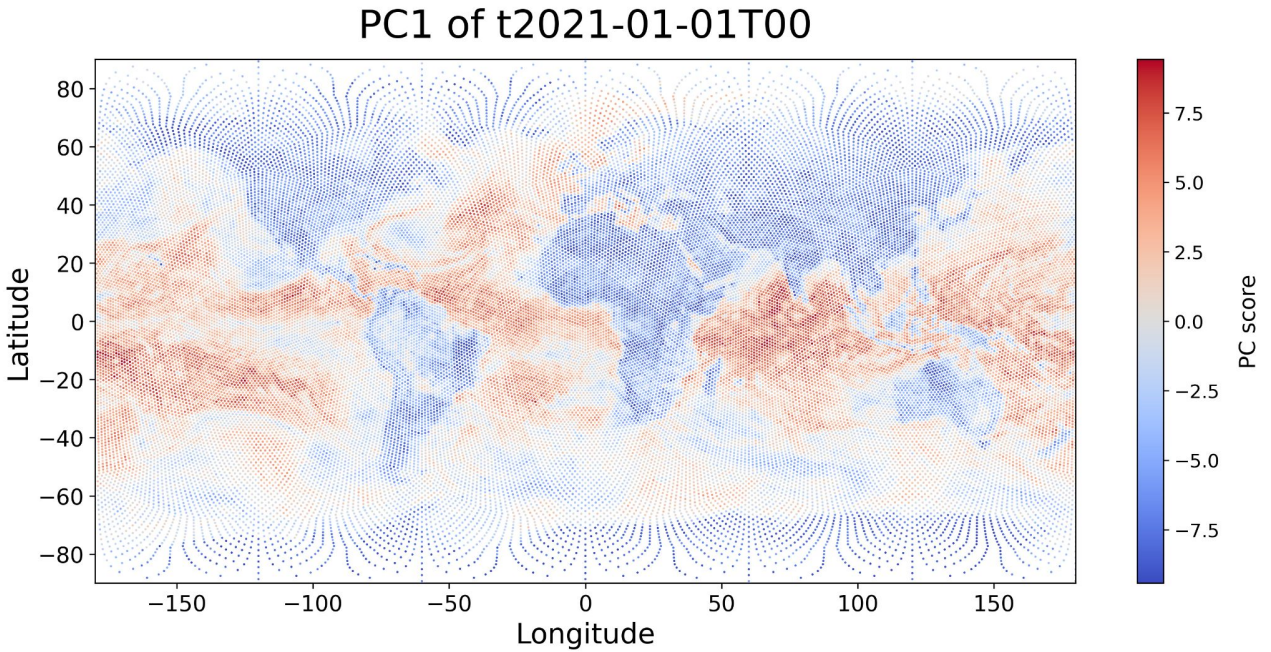


Stack all **node activations** and **timesteps**

Fit PCA to the 512 latent channel space

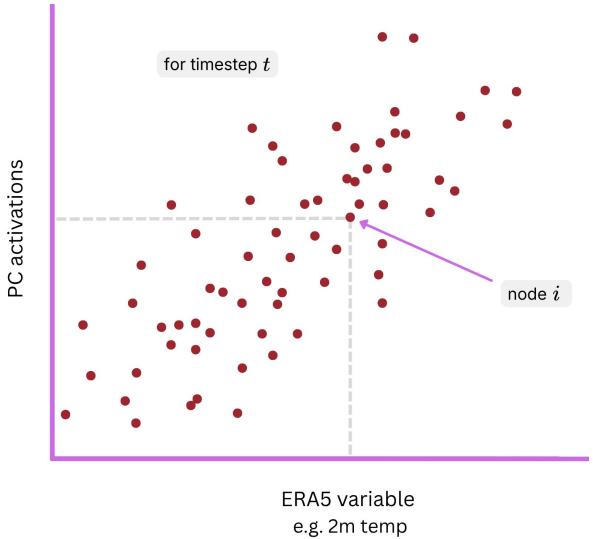


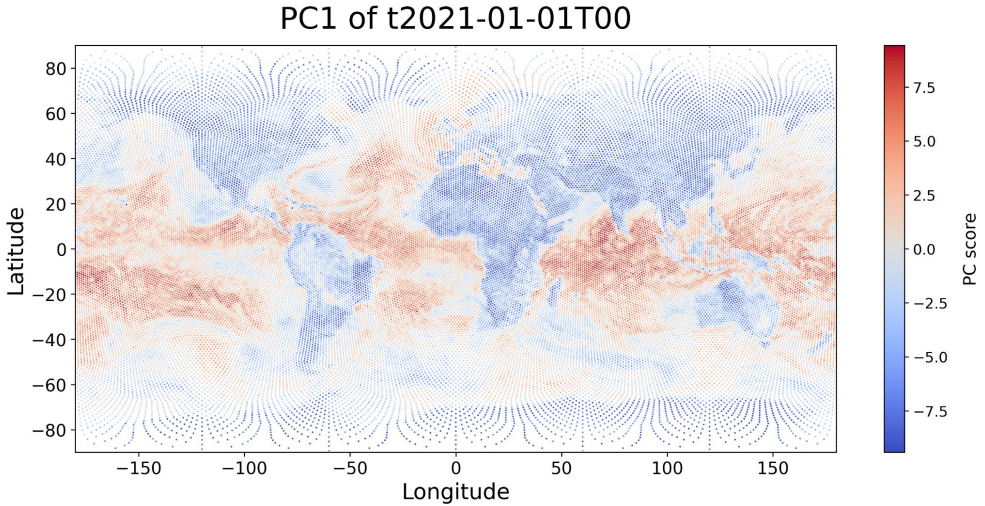
Fit PCA onto activations from 2019, 2020, apply it onto 2021



Map per PC global activations onto a specific day

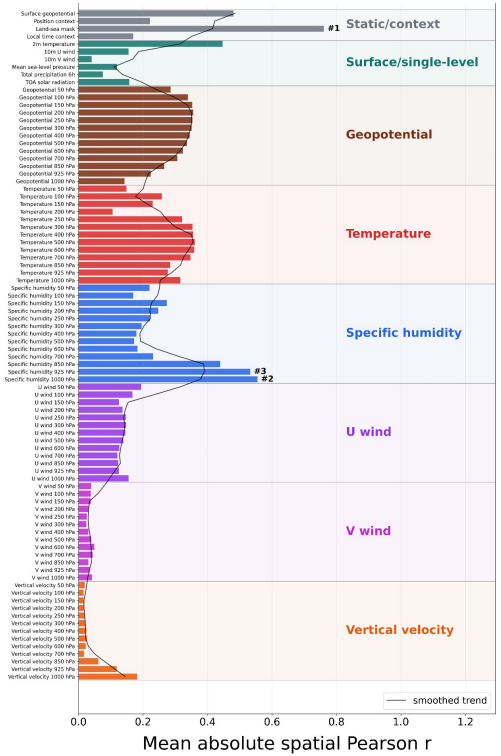
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

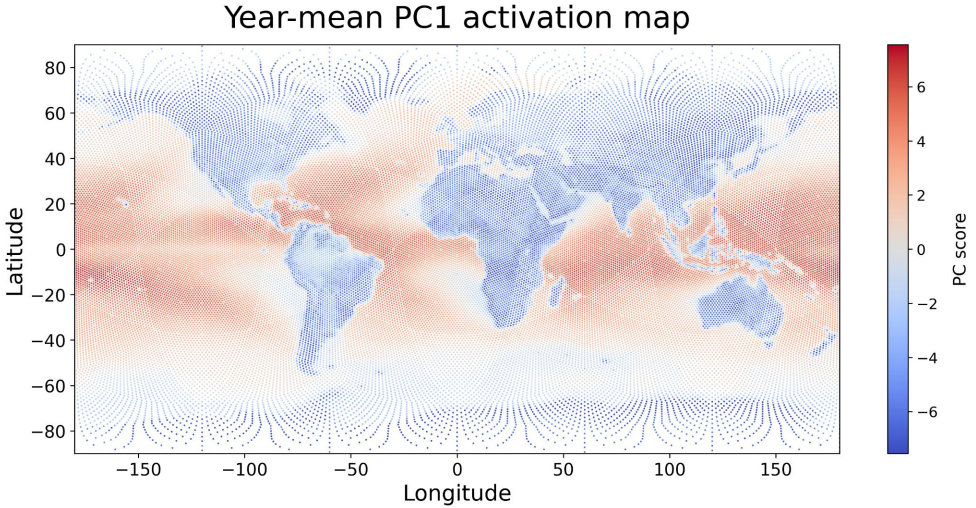




Correlate PC activations to ERA5 variables

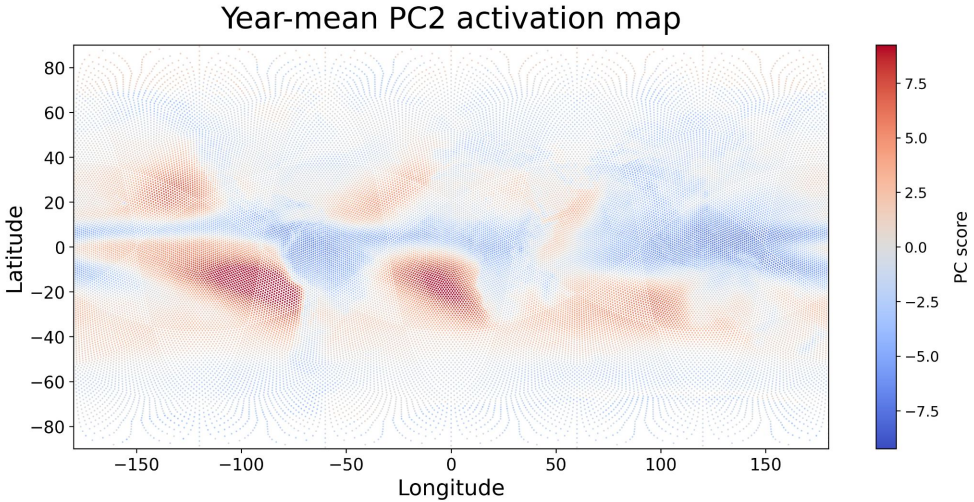
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

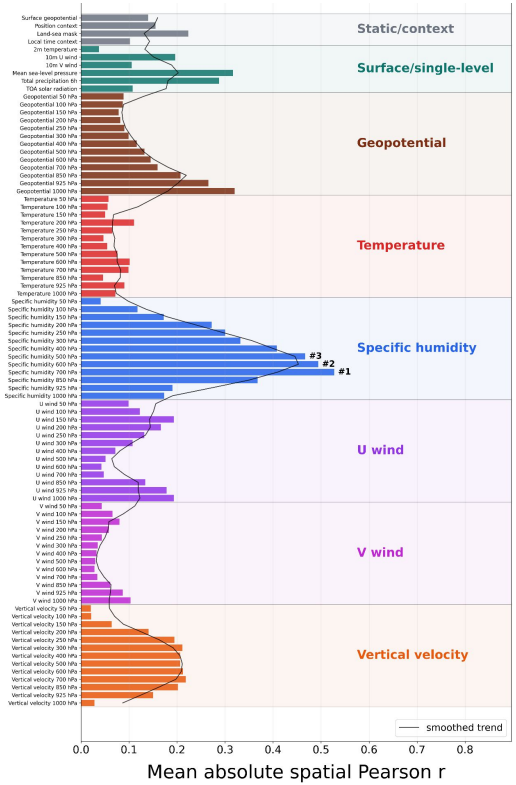




PC1 is strongly correlated to a land-sea mask, and specific humidity

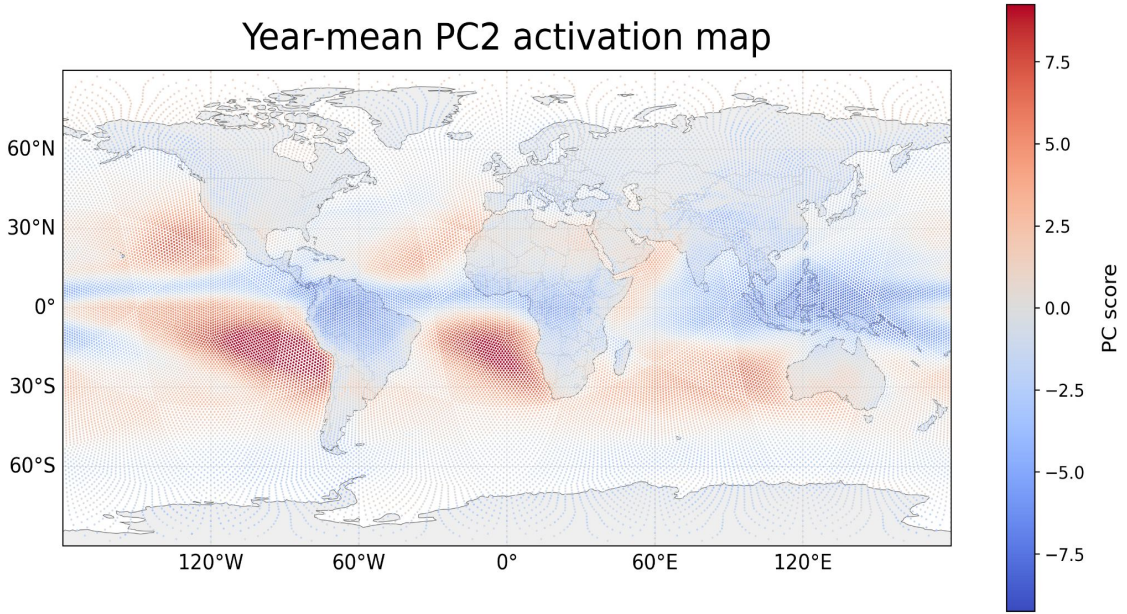
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY



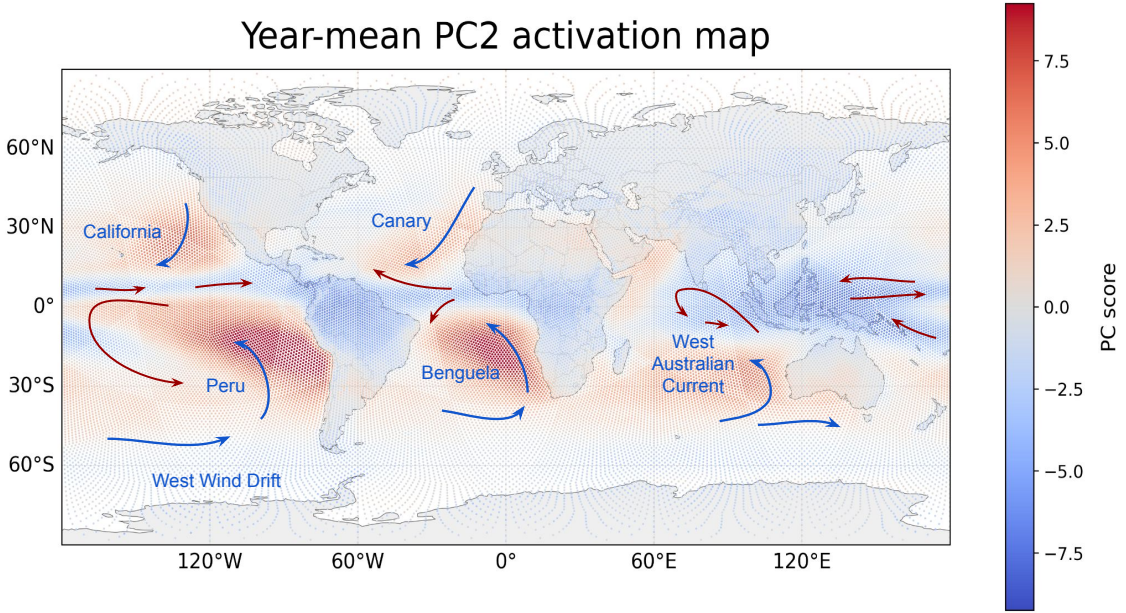


PC2 is most correlated to specific humidity

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

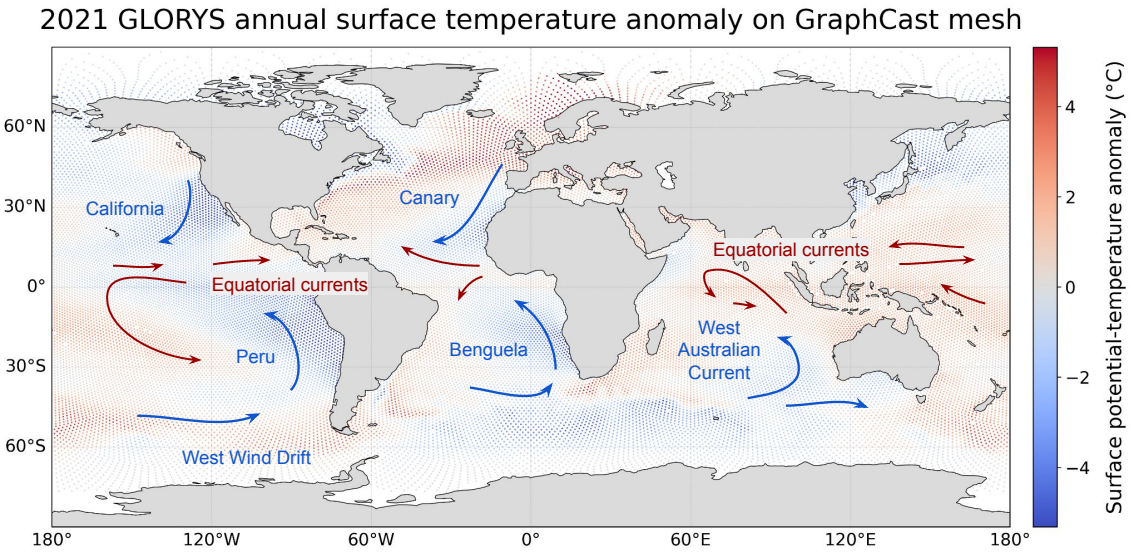


PC2 is also strongly aligned to cold/warm ocean currents

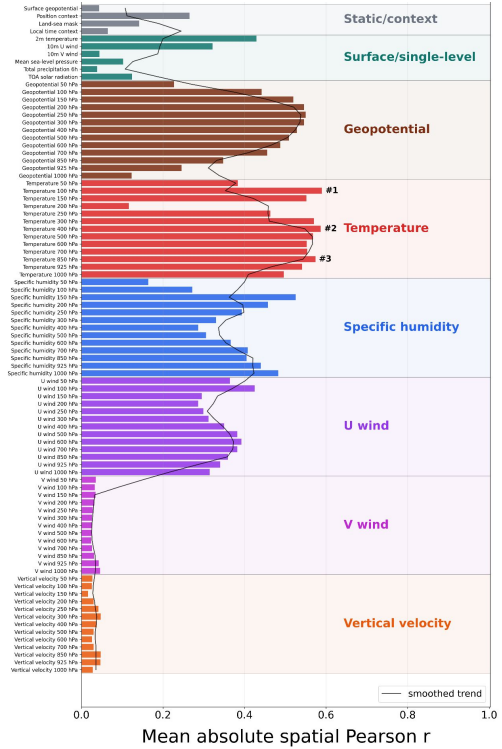


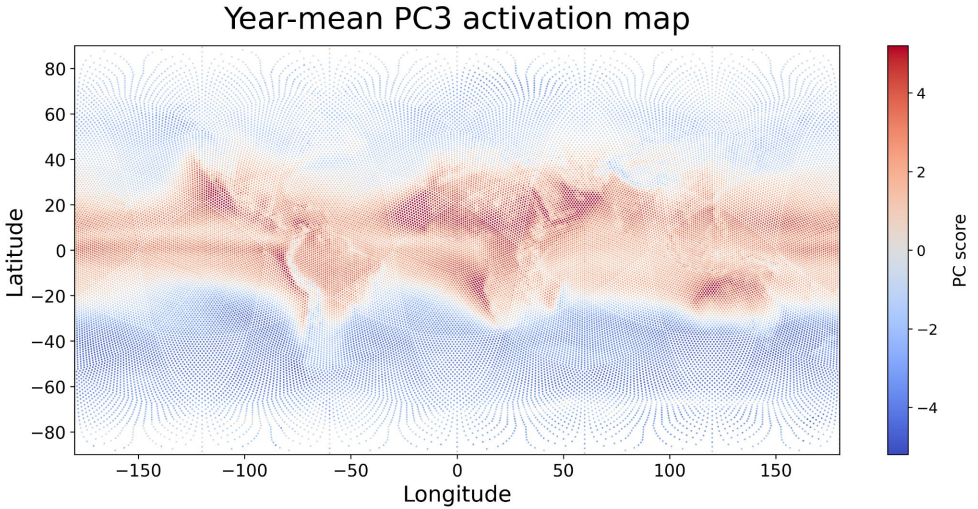
PC2 is also aligned to cold/warm ocean currents

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY



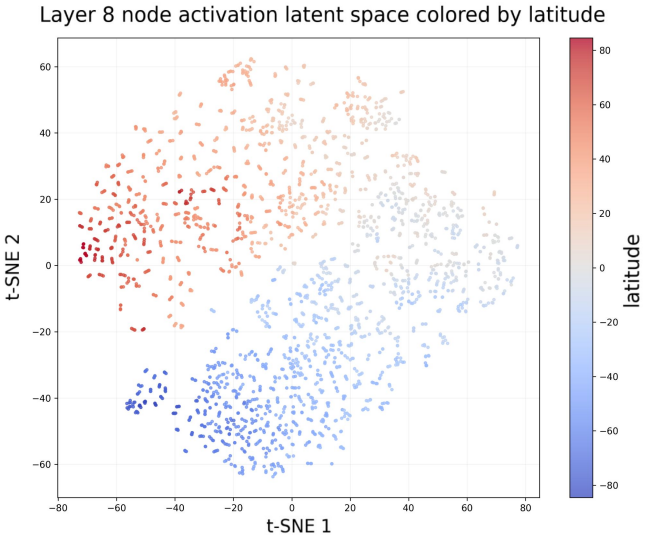
PC2 is also aligned to cold/warm ocean currents

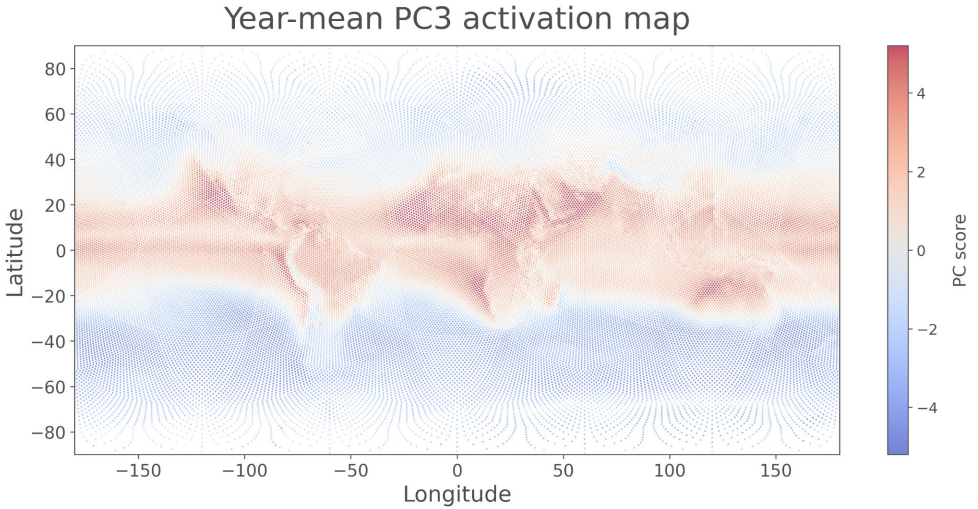




PC3 is most correlated to temperature

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

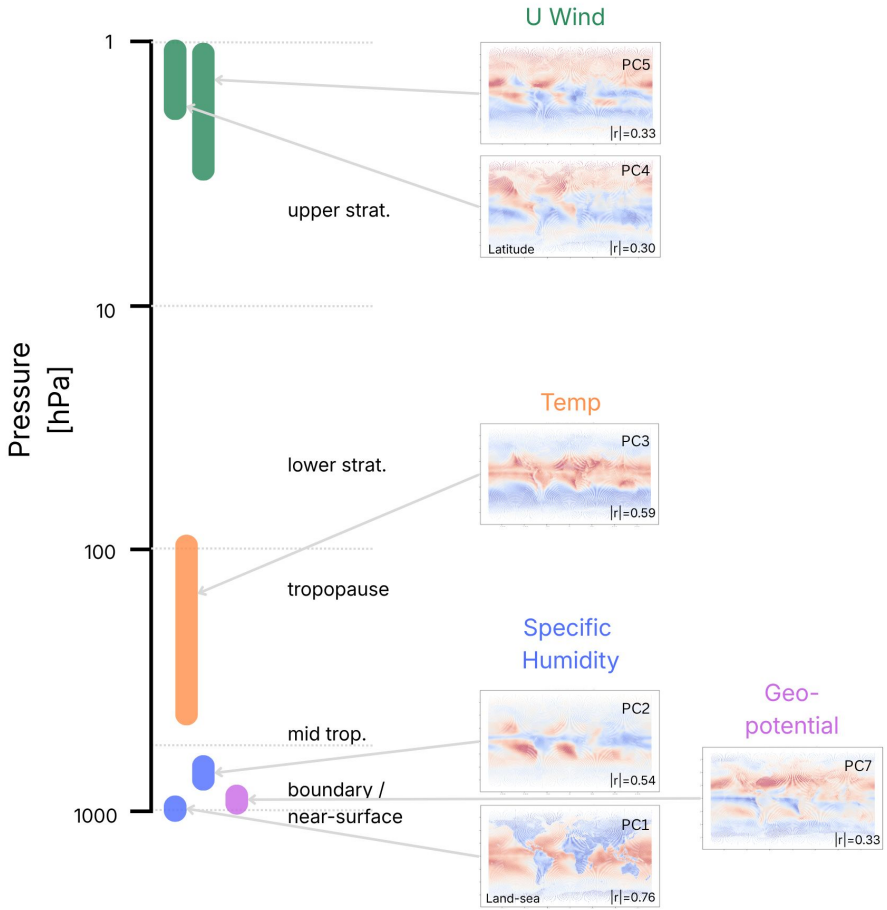




PC3 is most correlated to temperature

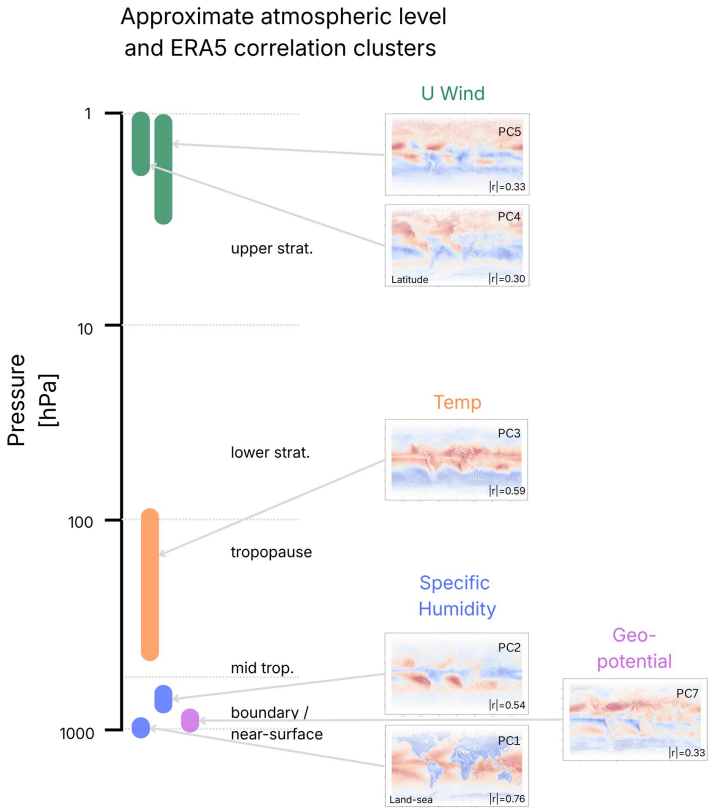
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

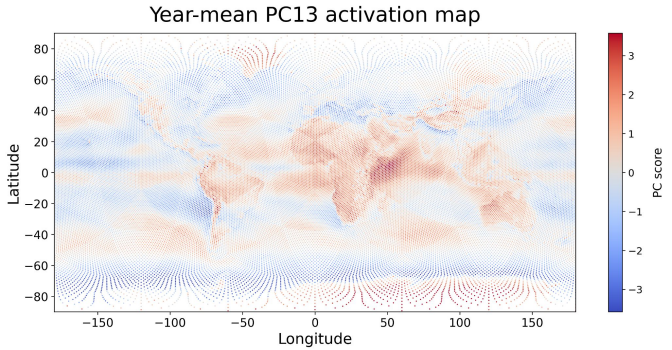
Approximate atmospheric level and ERA5 correlation clusters

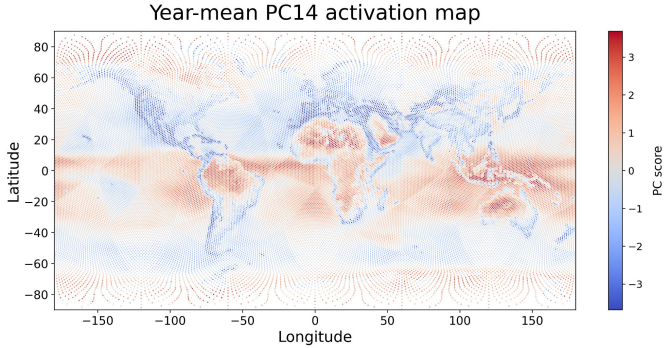


PCs are clustering at different atmospheric levels

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

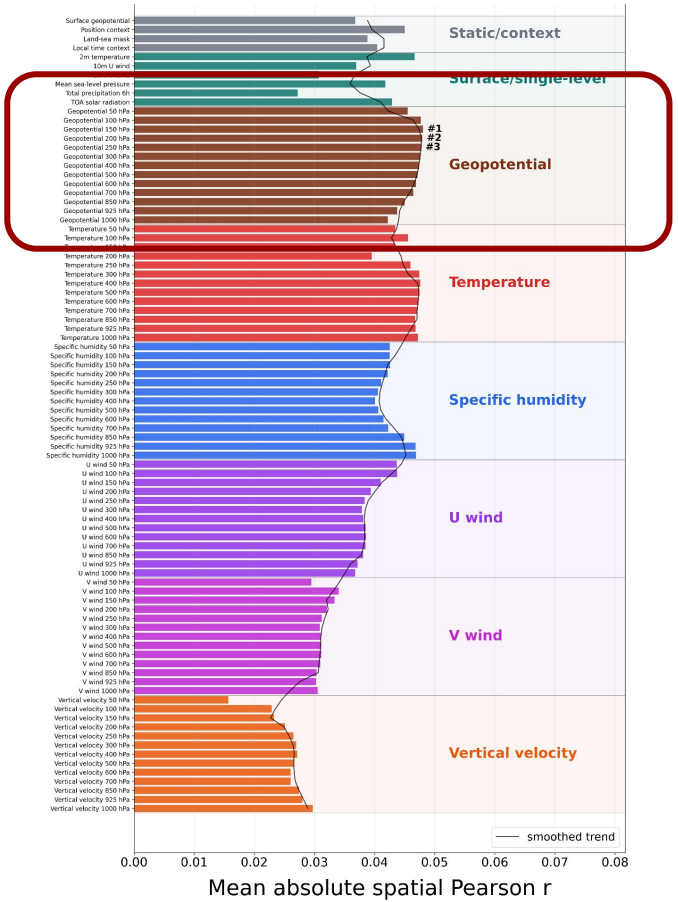






Later PC spatial maps show e.g. specific continents and coastlines

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

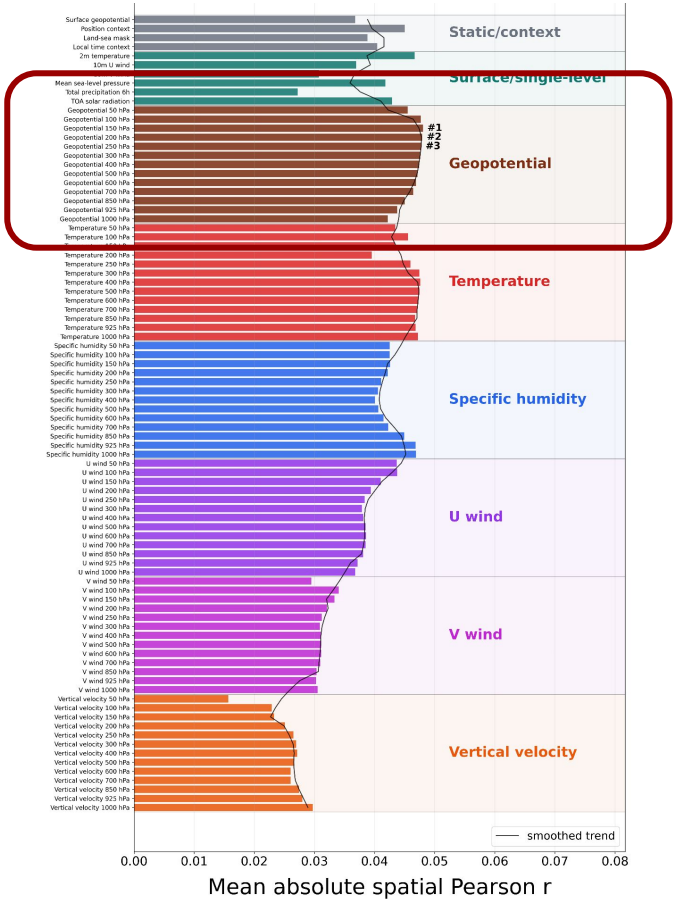


Mean |r| across ALL PCs:

Overall geopotential correlations best over all PCS.

Clear patterns across **variables and atmospheric levels.**

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY



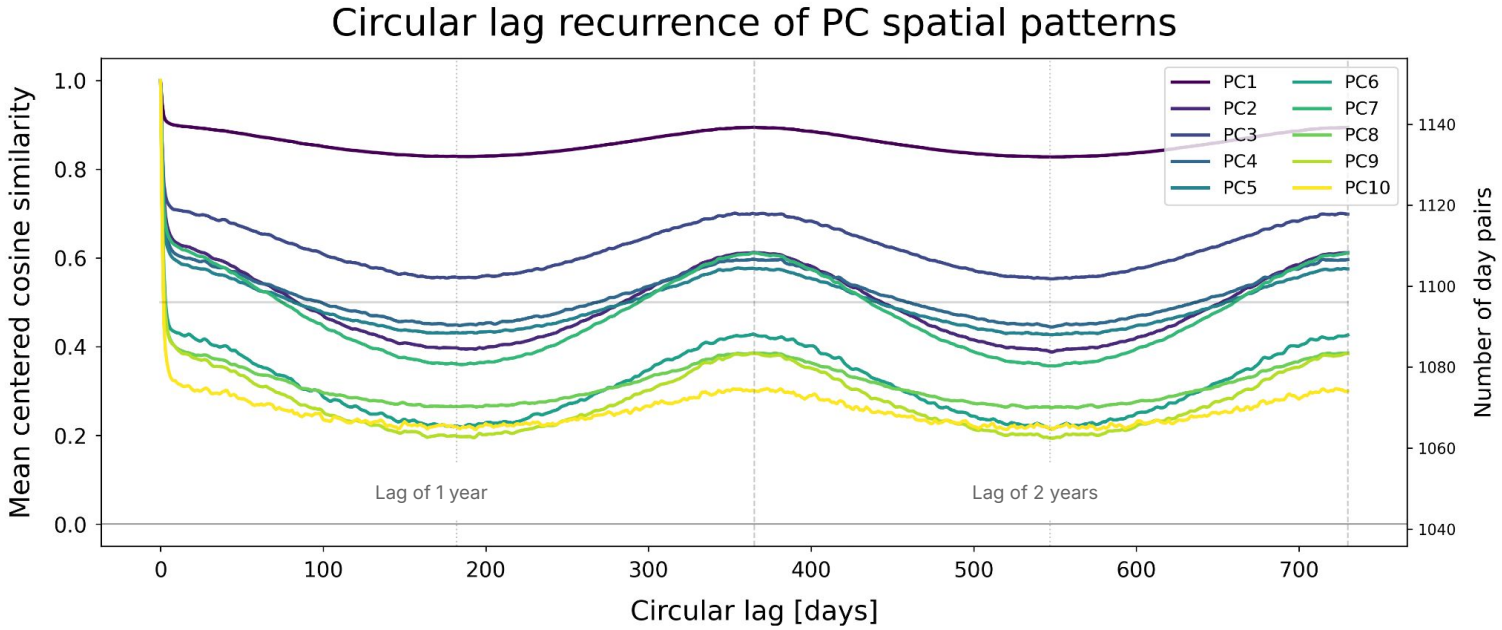
Mean |r| across ALL PCs:

Overall geopotential correlations best over all PCS.

Clear patterns across **variables and atmospheric levels.**

... but what about time?

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY



Despite time and space being flattened, PC activations show seasonal cycles

Does Graphcast learn interpretable physical features?

1. We find **strong correlations** between PC activations and specific ERA5 variables.
2. Provide a view of **relative structures** of its representational space.

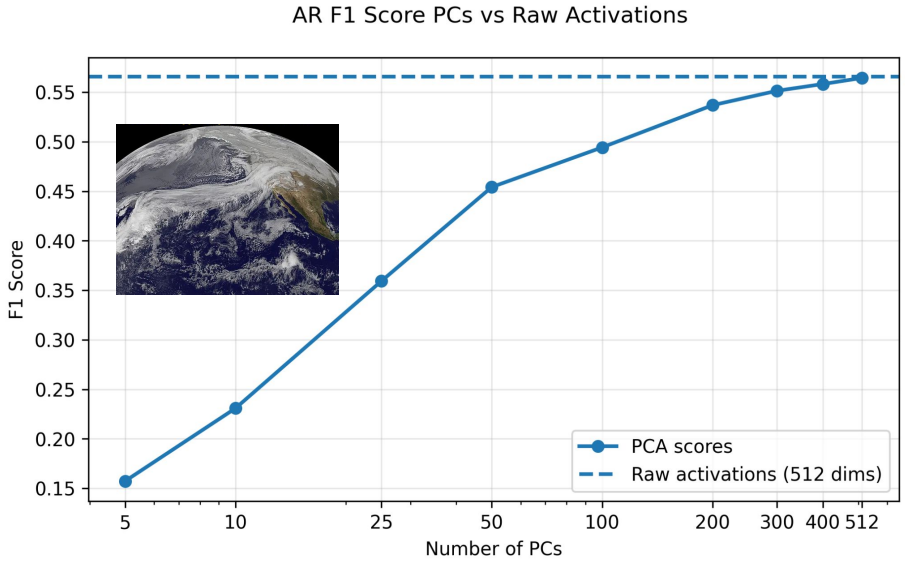
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

Workshop on Machine Learning for Earth System Modelling
24–26 August 2026 · Bonn, Germany

FINDING WEATHER EVENTS

MALIN BRAATZ, SABINE SCHOLLE | OSNABRÜCK UNIVERSITY

Logistic Regression for AR Detection

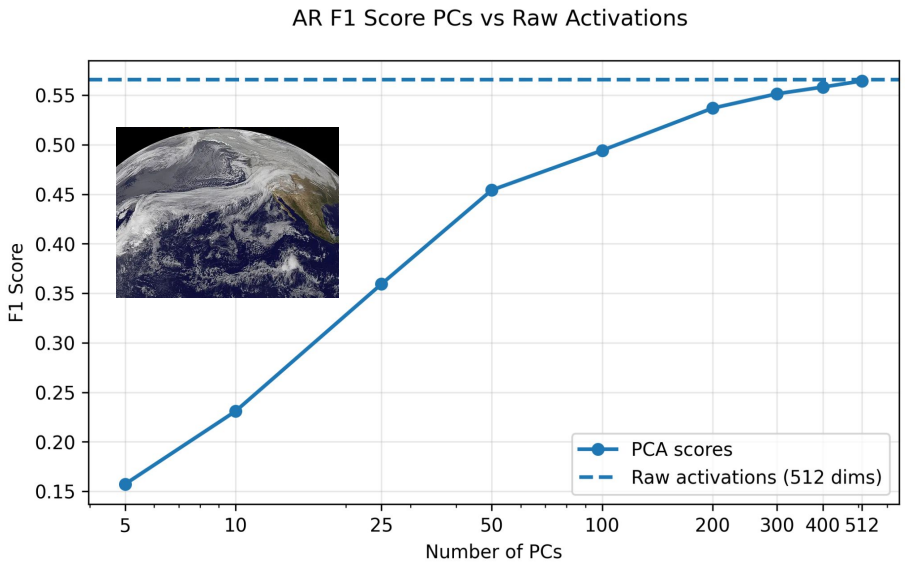


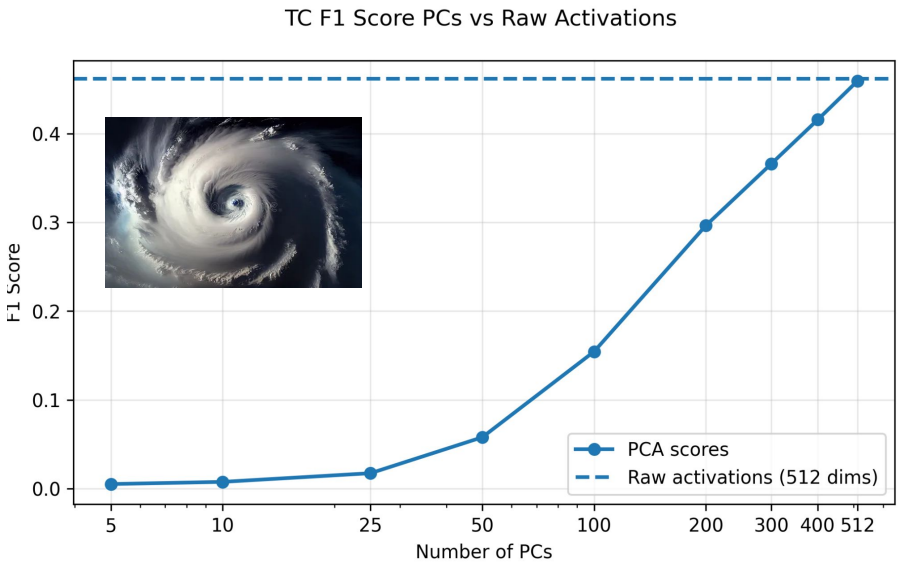
Trained using the ClimateNet expert-labeled AR and TC datasets

AR nodes are decodable from the latent space

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

Logistic Regression for AR/TC Detection

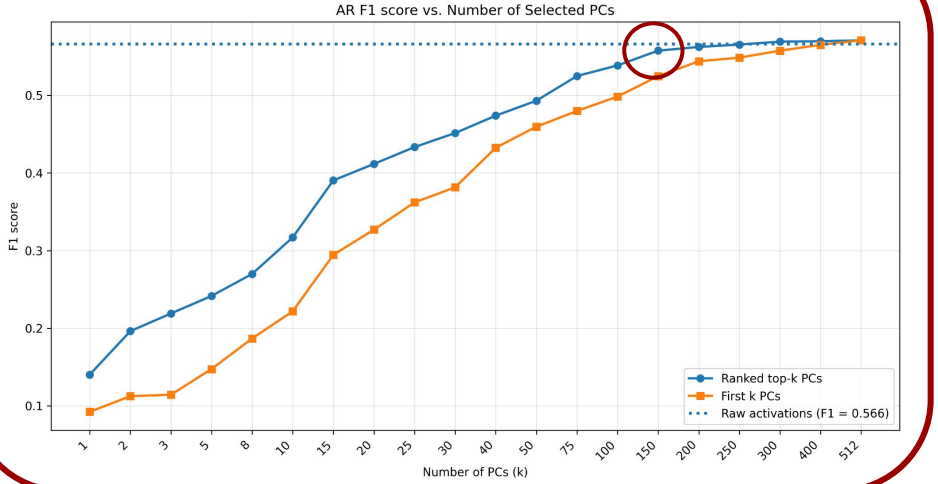


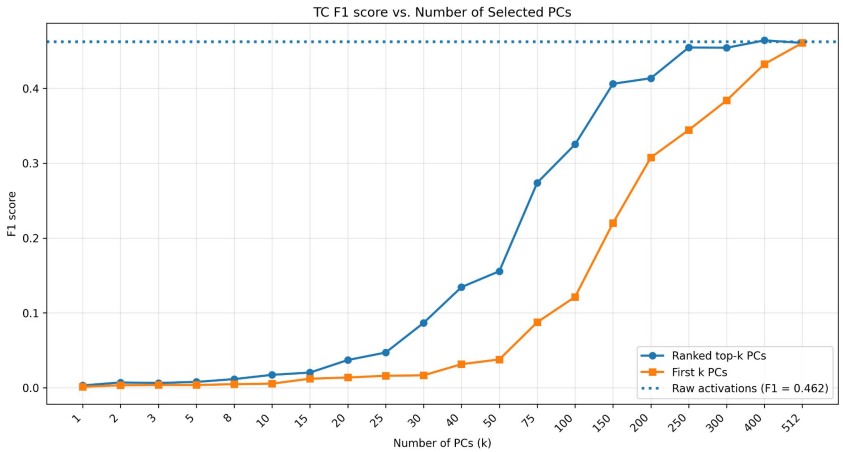


Trained using the ClimateNet expert-labeled AR and TC datasets

AR and TC nodes are decodable from the latent space

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY





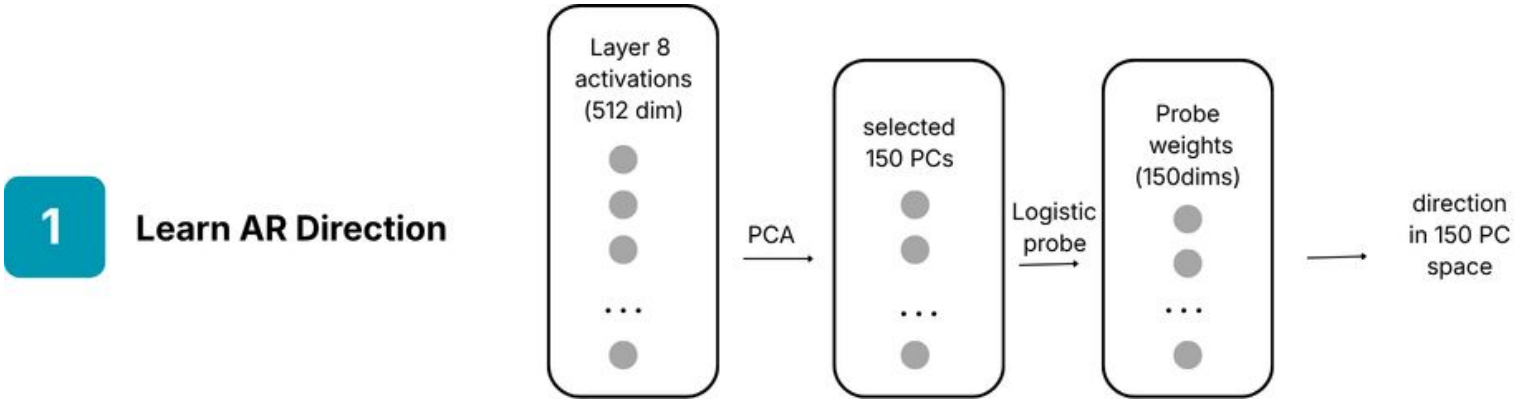
By selecting PCs ARs and TCs can be decoded from a substantially lower-dimensional subset

Workshop on Machine Learning for Earth System Modelling
24–26 August 2026 · Bonn, Germany

LATENT PERTURBATION

MALIN BRAATZ, SABINE SCHOLLE | OSNABRÜCK UNIVERSITY





Perturbation methodology

1 Learn AR Direction → 2 Select & Pertub Nodes → 3 Roll out Forecast → 4 Measure Effect

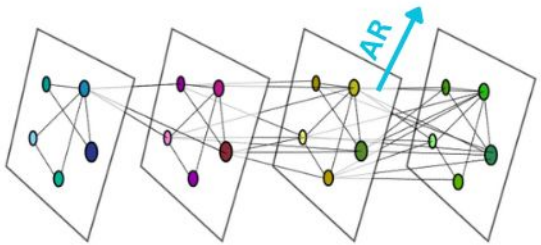
2 **Select & Pertub Nodes**

For each mesh node i at layer 8
select nodes $P_{AR}(i) \geq 0.9$

Pertub selected nodes:

$$x_i' = x_i + \gamma \cdot P_{AR}(i) \cdot d_{AR}$$

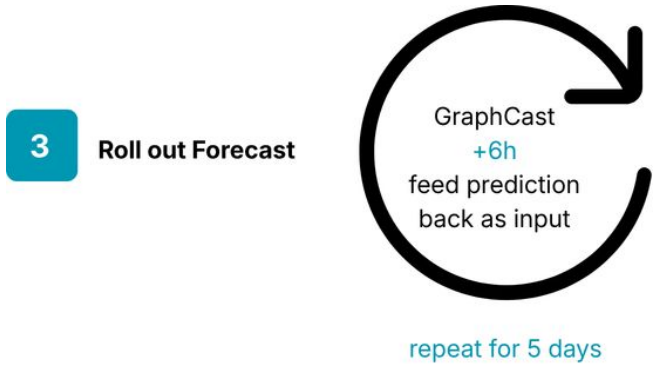
pertubed activation
original activation
pertubation strength
confidence
direction



Detect AR notes & perturbate them

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY





Rollout forecast



4 Measure Effect

Compare IVT of perturbed forecast against baseline ($\gamma = 0$) to measure whether the AR signal changed

Compare IVT:

$$\Delta\mathrm{IVT}(t) = \mathrm{IVT}_{\gamma}(t) - \mathrm{IVT}_{\gamma=0}(t)$$

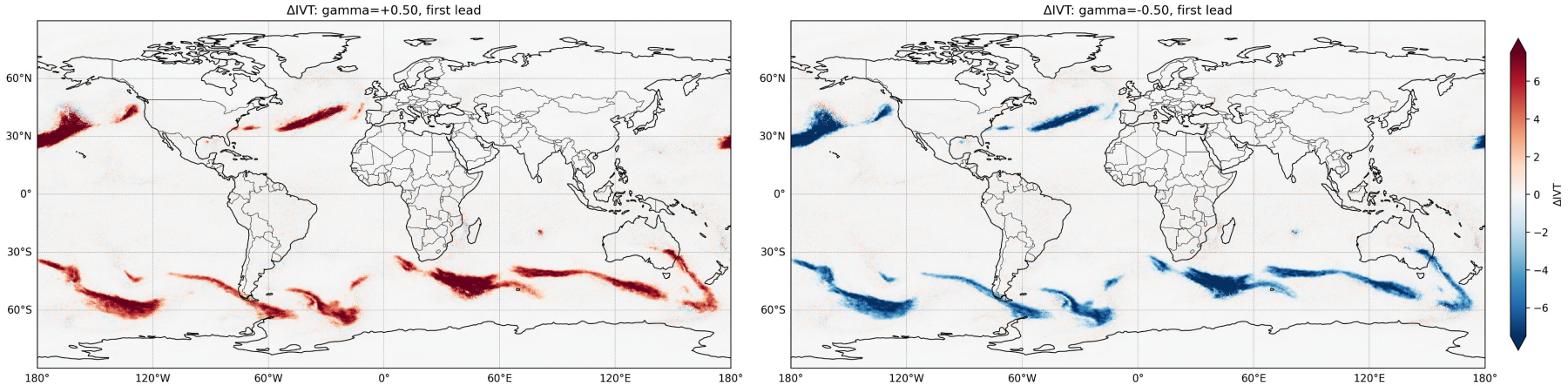
where

$$\mathrm{IVT} = \sqrt{\left(-\frac{1}{g}\int_{p_B}^{p_T} qu\,dp\right)^2 + \left(-\frac{1}{g}\int_{p_B}^{p_T} qv\,dp\right)^2}$$

u, v = zonal and meridional winds
q = specific humidity
g = gravitational acceleration
p_B, p_T = bottom and top pressure

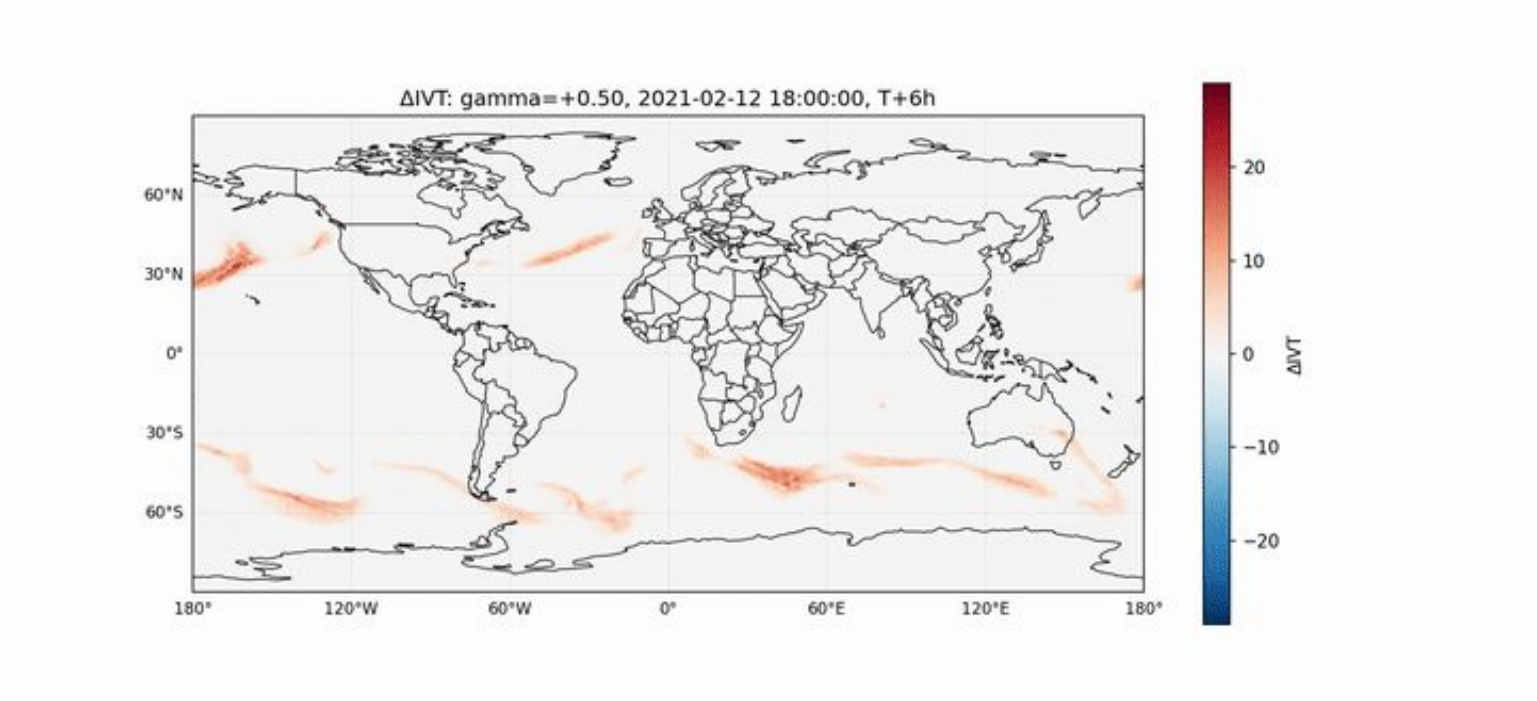
Measure the result of the perturbation

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

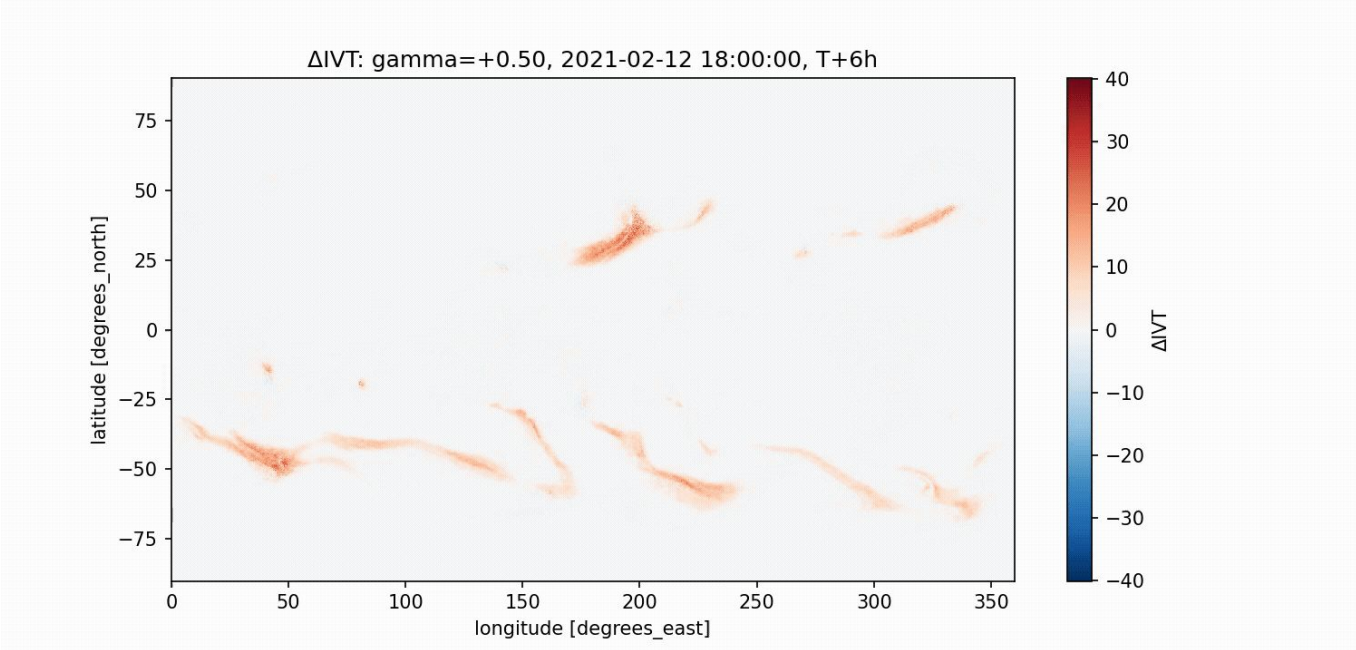


Perturbing the AR signal in latent space leads to an increased AR signal in the forecast

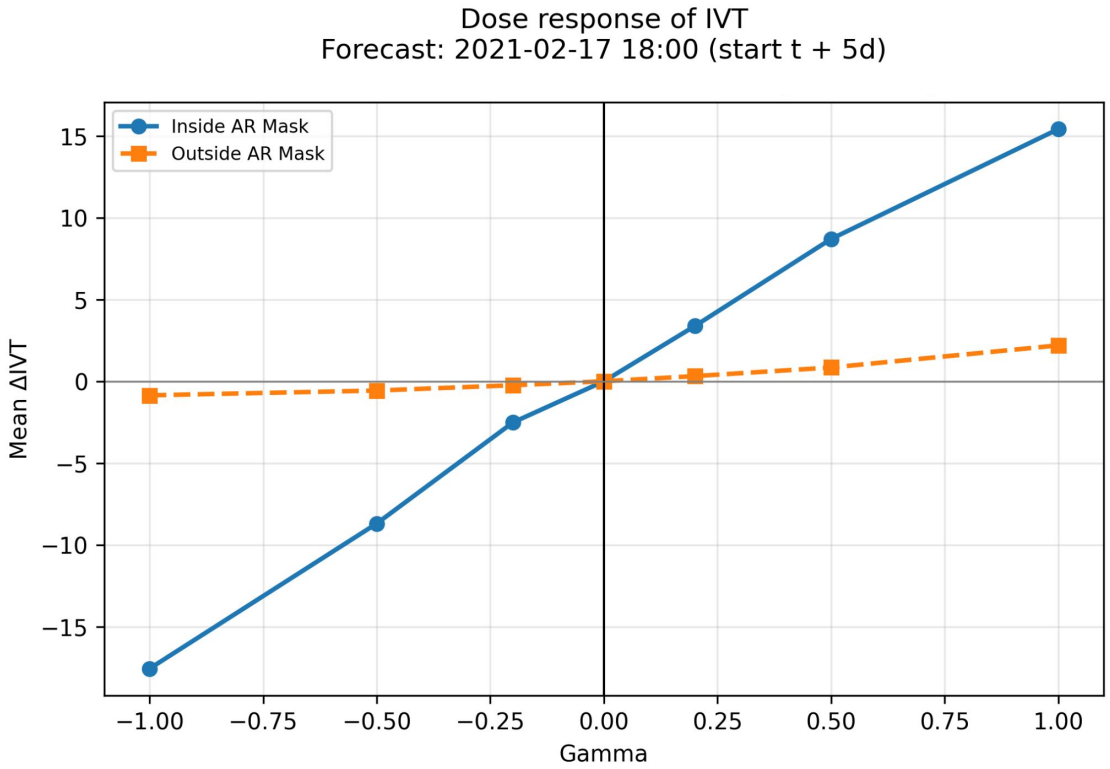
MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY



Perturbing the AR signal in latent space leads to an increased AR signal in the forecast



Perturbing the AR signal in latent space leads to an increased AR signal in the forecast



The perturbation only affects the expected AR regions

MALIN BRAATZ, SABINE SCHOLLE
OSNABRÜCK UNIVERSITY

Takeaways

The **first PCs** of GC's latent space are **strongly correlated to specific ERA5 variables**, with **land-sea boundaries** and **near-surface humidity** most strongly informing the first PCs, followed by **tropospheric temperatures** and **upper stratospheric wind fields**.

Complex weather events such as **atmospheric rivers and tropical cyclones** can be **decoded** in GC latent space, and **perturbed** using a **subset of PCs.**

Future

Tensor decomposition and SINDy Autoencoder for further physical event identification.

Thank you

Prof. Dr. Pascal Nieters
Robin Rawiel MSc.
Dr. Dwaipayan Chatterjee

UNIVERSITÄT OSNABRÜCK

Resources

Hakim, G. J., & Masanam, S. (2024). Dynamical tests of a deep learning weather prediction model. *Artificial Intelligence for the Earth Systems*, *3*(3). https://doi.org/10.1175/aies-d-23-0090.1

Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirnsberger, P., Fortunato, M., Alet, F., Ravuri, S., Ewalds, T., Eaton-Rosen, Z., Hu, W., Merose, A., Hoyer, S., Holland, G., Vinyals, O., Stott, J., Pritzel, A., Mohamed, S., & Battaglia, P. (2022, December 24). *GraphCast: Learning skillful medium-range global weather forecasting*. arXiv.org. https://arxiv.org/abs/2212.12794

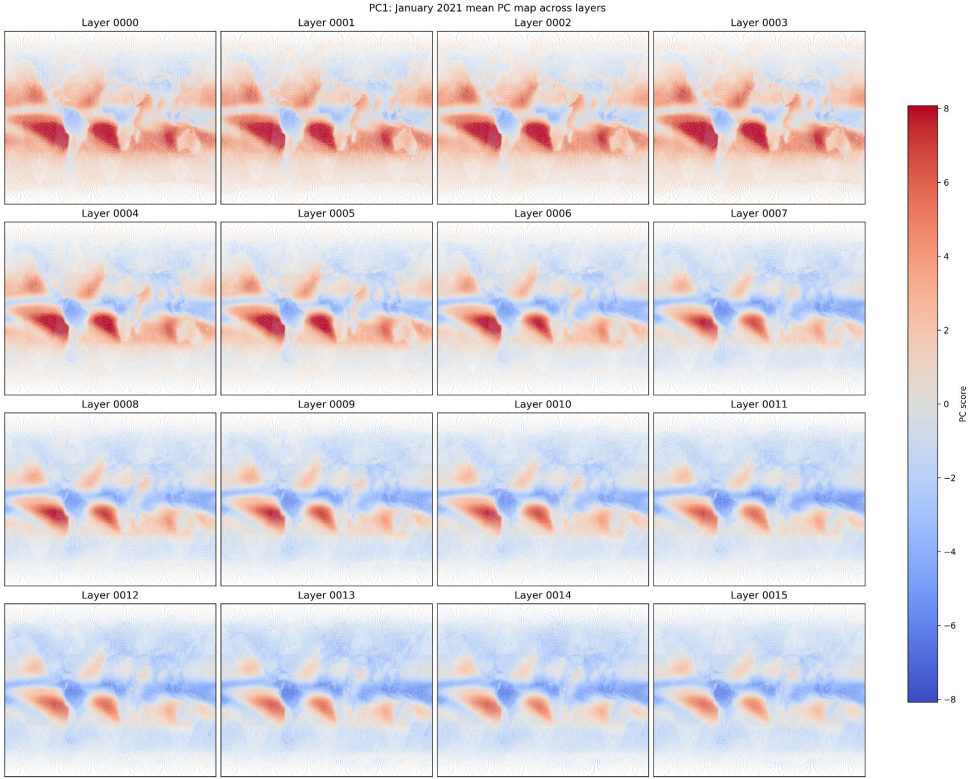
MacMillan, T., & Ouellette, N. T. (2025, December 30). *Towards mechanistic understanding in a data-driven weather model: internal activations reveal interpretable physical features*. arXiv.org. https://arxiv.org/abs/2512.24440v1

Prabhat, Kashinath, K., Mudigonda, M., Kim, S., Kapp-Schwoerer, L., Graubner, A., Karaismailoglu, E., Von Kleist, L., Kurth, T., Greiner, A., Mahesh, A., Yang, K., Lewis, C., Chen, J., Lou, A., Chandran, S., Toms, B., Chapman, W., Dagon, K., . . . Collins, W. (2021). ClimateNet: an expert-labeled open dataset and deep learning architecture for enabling high-precision analyses of extreme weather. *Geoscientific Model Development*, *14*(1), 107–124. https://doi.org/10.5194/gmd-14-107-2021

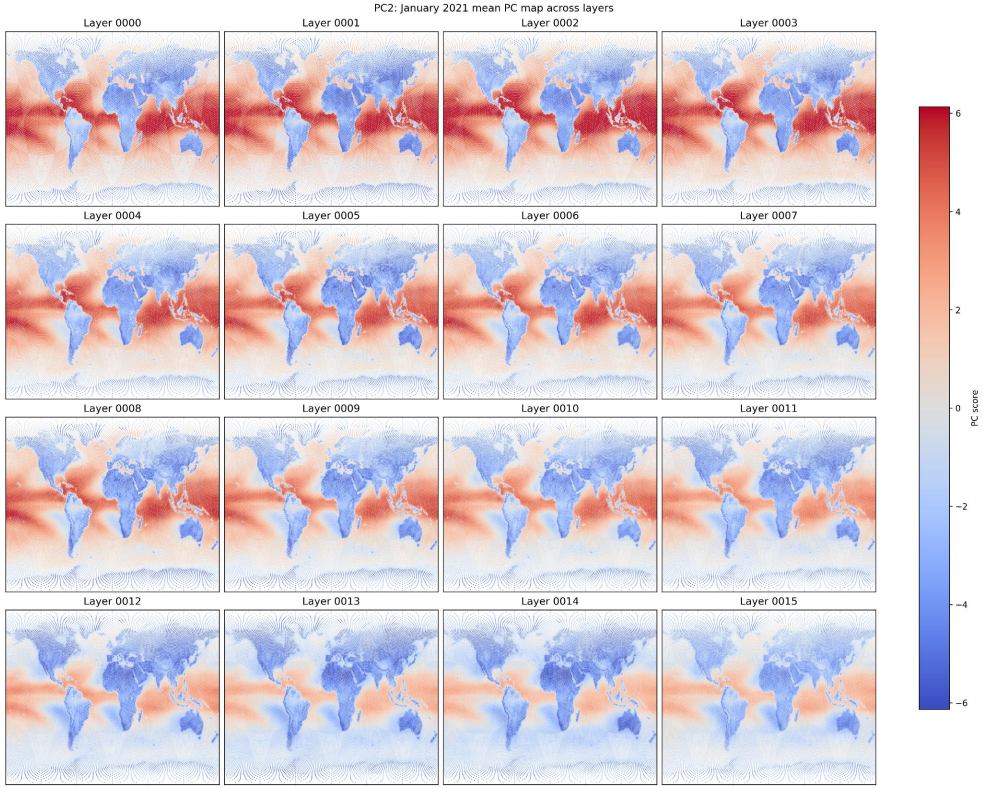
Tempest, K., I., Beylich, M., & Craig, G. C. (2026, April 22). *Mechanistic Interpretability Tool for AI weather models*. arXiv.org. https://arxiv.org/abs/2604.20467v1

APPENDIX

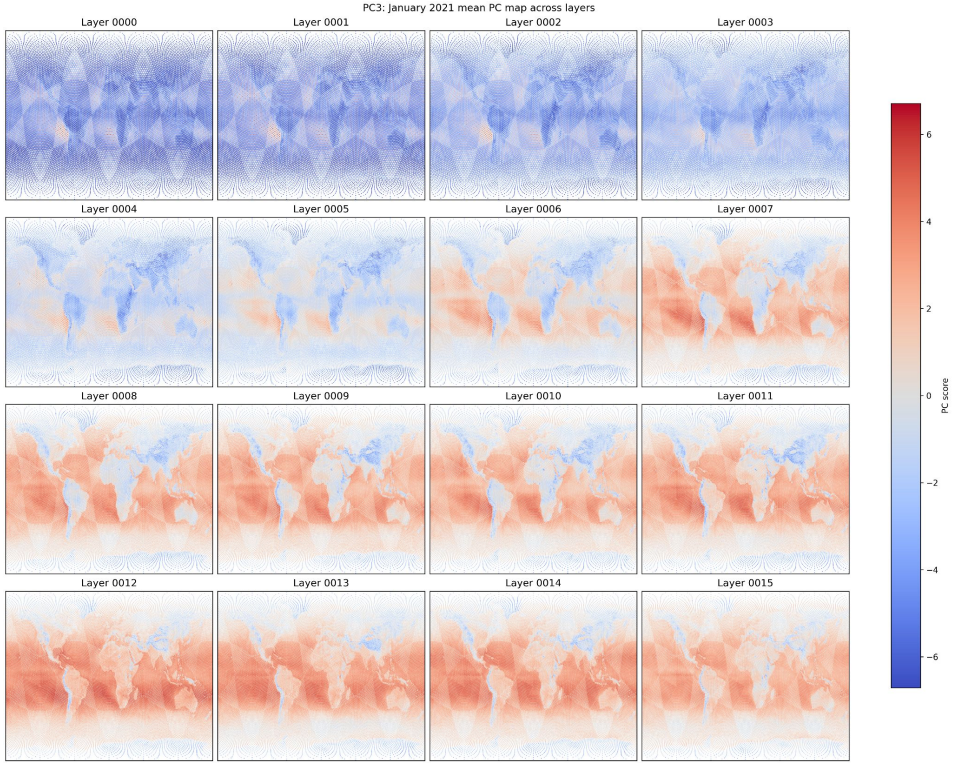
MALIN BRAATZ, SABINE SCHOLLE | OSNABRÜCK UNIVERSITY



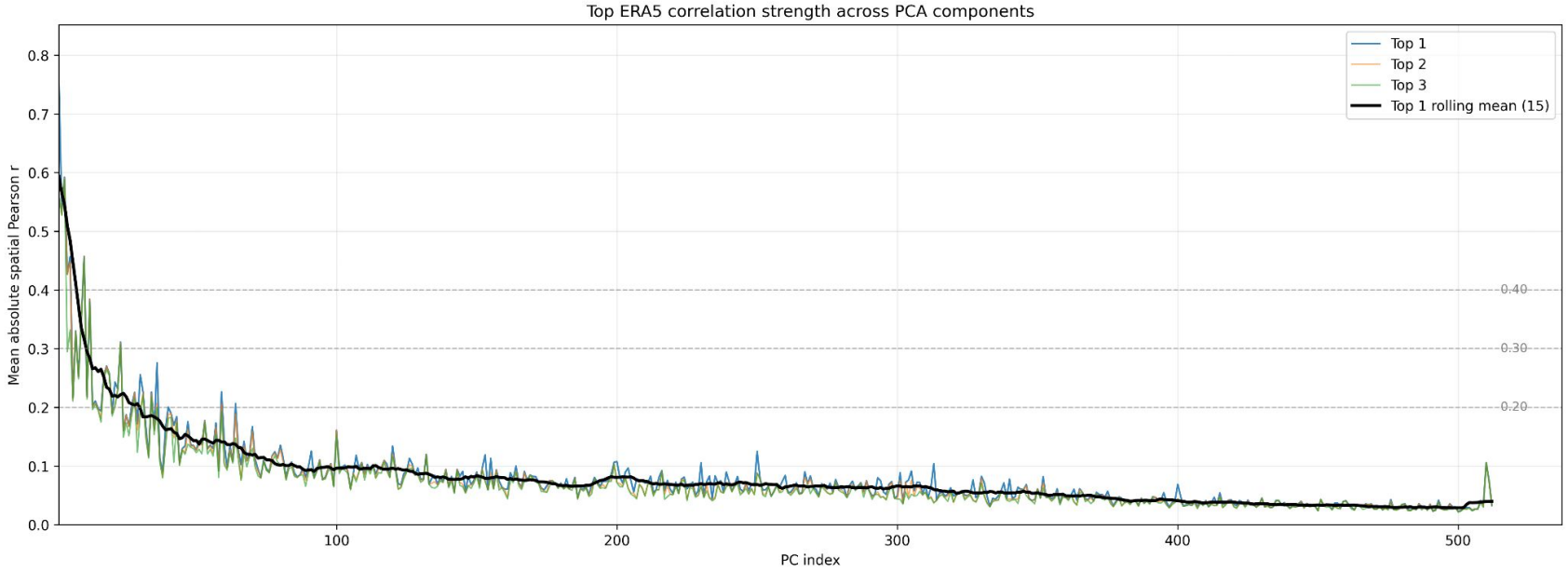
PC1 fitted and mapped on all layers of 2019

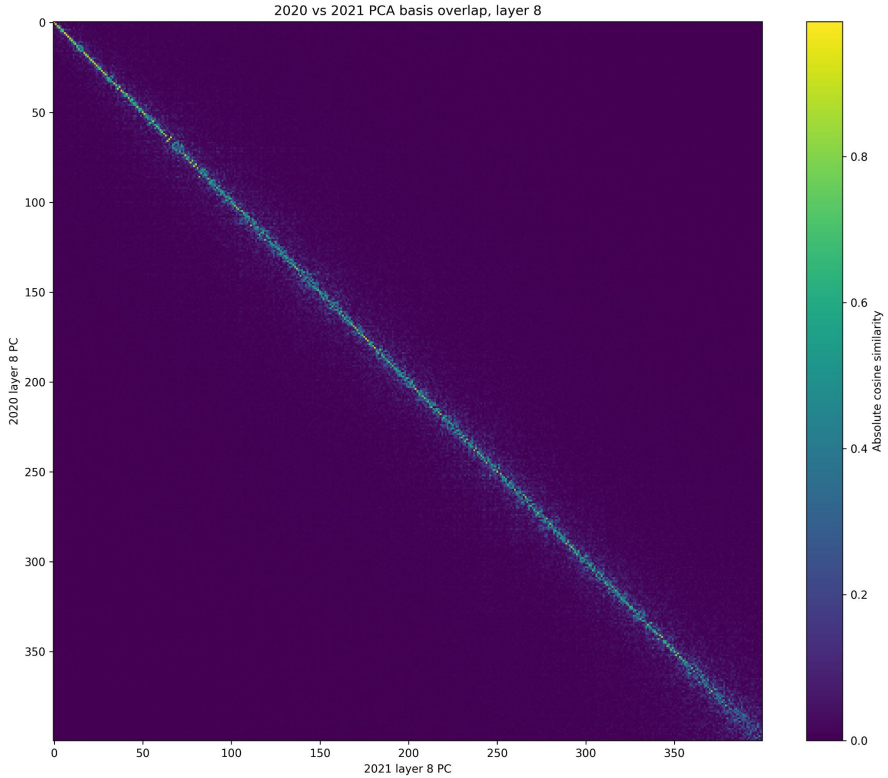


PC2 fitted and mapped on all layers of 2019

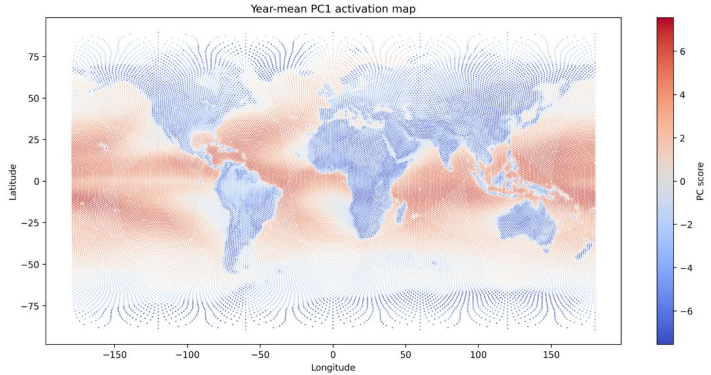


PC3 fitted and mapped on all layers of 2019


Top ERA5 correlation strength across PCA components
Top 1
Top 2
Top 3
Top 1 rolling mean (15)
Mean absolute spatial Pearson r
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.40
0.30
0.20
100
200
300
400
500
PC index


2020 vs 2021 PCA basis overlap, layer 8
2020 layer 8 PC
2021 layer 8 PC
Absolute cosine similarity
0
50
100
150
200
250
300
350
0.0
0.2
0.4
0.6
0.8

PC1: year-mean activation map



Top ERA5 matches

#1 Land-sea mask

mean |r| = 0.761

single-level / static context

#2 Specific humidity, lev35

mean |r| = 0.562

975 hPa; boundary layer / near-surface

#3 Specific humidity, lev34

mean |r| = 0.560

950 hPa; boundary layer / near-surface

#4 Specific humidity, lev36

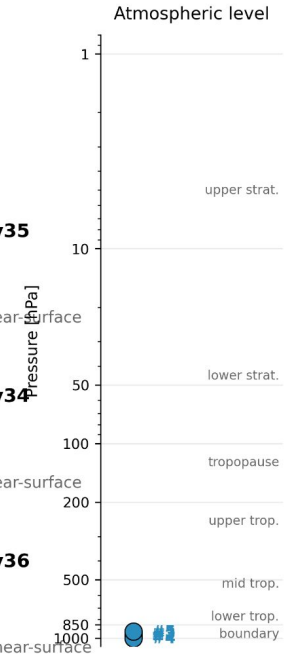
mean |r| = 0.555

1000 hPa; boundary layer / near-surface

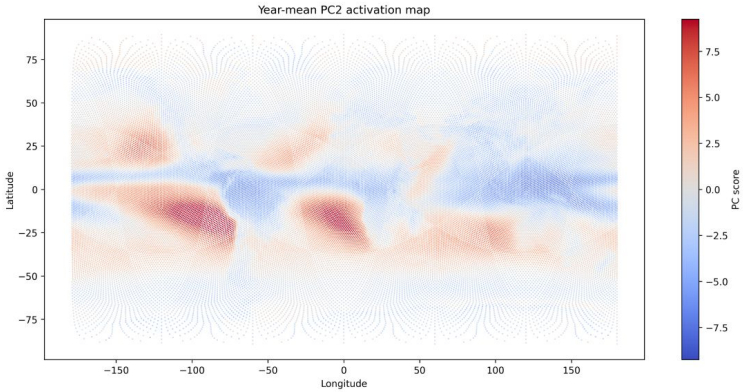
#5 Specific humidity, lev33

mean |r| = 0.533

925 hPa; boundary layer / near-surface



PC2: year-mean activation map



Top ERA5 matches

#1 Specific humidity, lev26

mean |r| = 0.539

750 hPa; lower troposphere

#2 Specific humidity, lev27

mean |r| = 0.528

775 hPa; lower troposphere

#3 Specific humidity, lev25

mean |r| = 0.527

700 hPa; lower troposphere

#4 Specific humidity, lev24

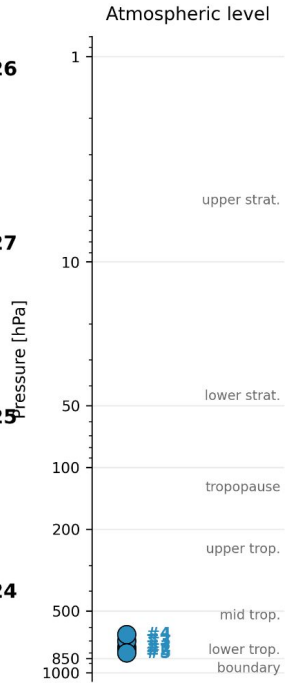
mean |r| = 0.506

650 hPa; mid troposphere

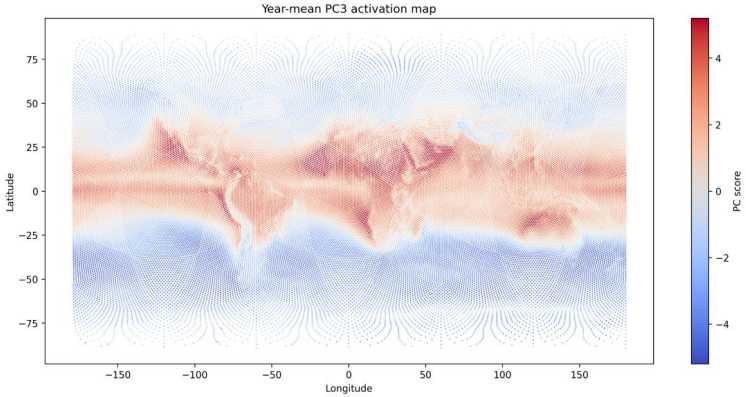
#5 Specific humidity, lev28

mean |r| = 0.502

800 hPa; lower troposphere



PC3: year-mean activation map



Top ERA5 matches

#1 Temperature, lev11

mean |r| = 0.592

125 hPa; tropopause / upper troposphere

#2 Temperature, lev10

mean |r| = 0.590

100 hPa; tropopause / upper troposphere

#3 Temperature, lev18

mean |r| = 0.588

350 hPa; upper troposphere

#4 Temperature, lev19

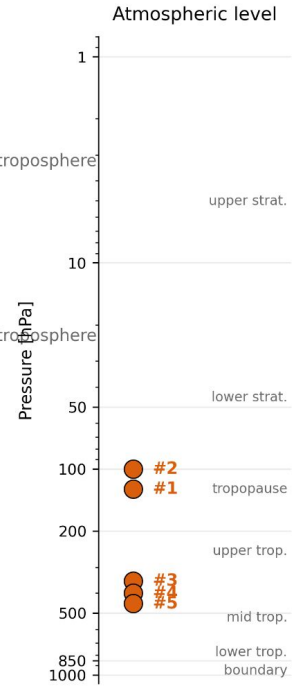
mean |r| = 0.587

400 hPa; mid troposphere

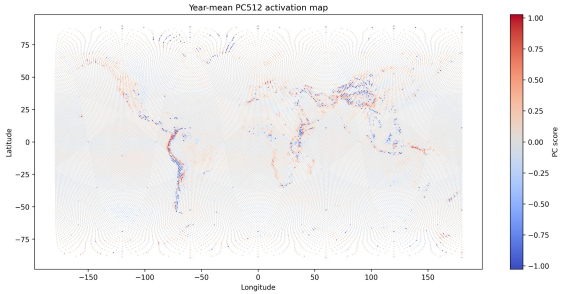
#5 Temperature, lev20

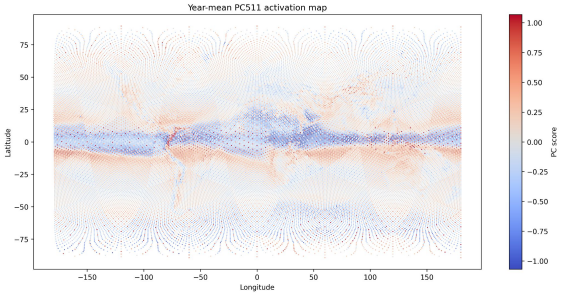
mean |r| = 0.578

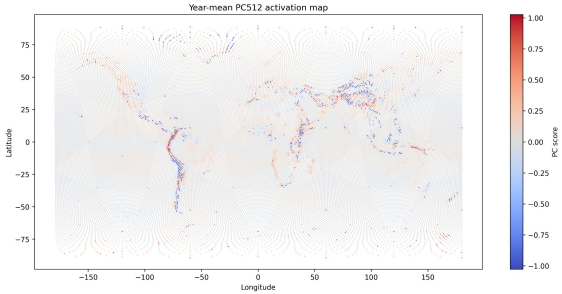
450 hPa; mid troposphere



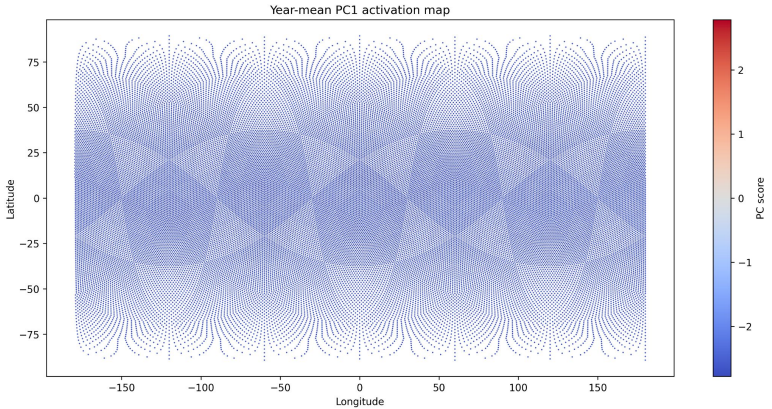
The last PCs

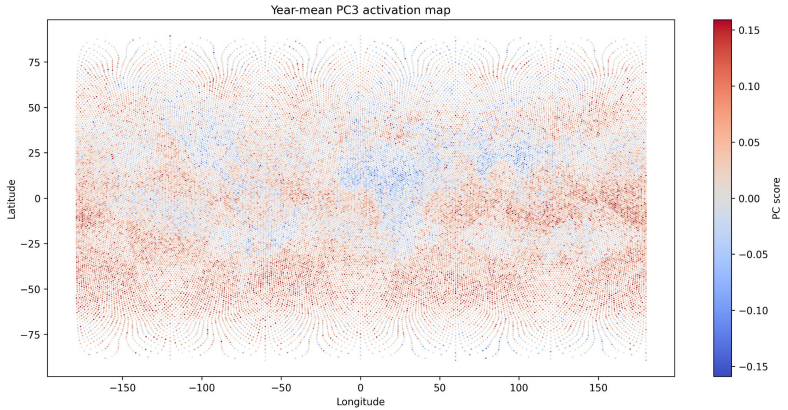

Year-mean PC512 activation map
Latitude
Longitude
PC score


Year-mean PC511 activation map
Latitude
Longitude
PC score


Year-mean PC512 activation map
Latitude
Longitude
PC score

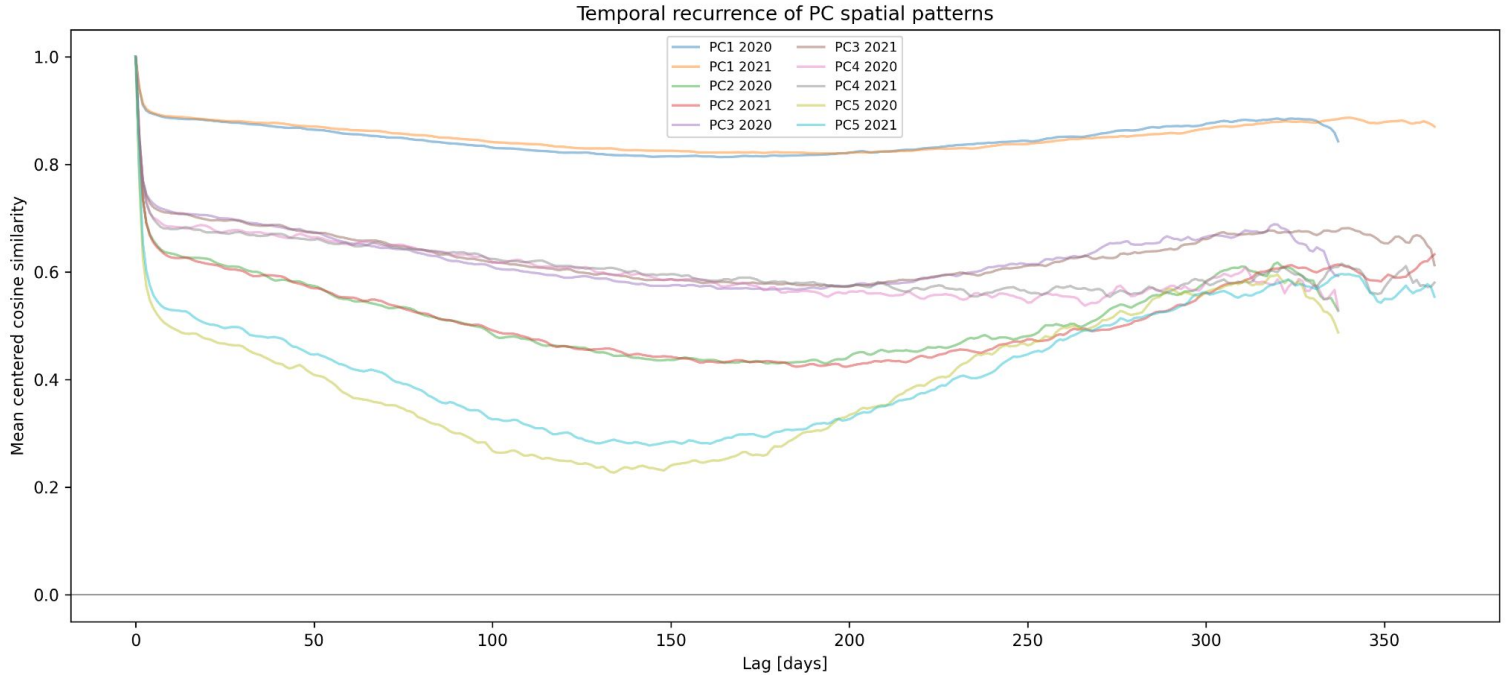
Scrambled PCs





lev00 = 1 hPa upper stratosphere / mesosphere
lev01 = 2 hPa upper stratosphere / mesosphere
lev02 = 3 hPa upper stratosphere / mesosphere
lev03 = 5 hPa upper stratosphere
lev04 = 7 hPa upper stratosphere
lev05 = 10 hPa upper stratosphere
lev06 = 20 hPa middle stratosphere
lev07 = 30 hPa middle stratosphere
lev08 = 50 hPa lower stratosphere
lev09 = 70 hPa lower stratosphere
lev10 = 100 hPa lower stratosphere / tropopause region
lev11 = 125 hPa tropopause region / upper troposphere
lev12 = 150 hPa upper troposphere / tropopause region
lev13 = 175 hPa upper troposphere
lev14 = 200 hPa upper troposphere
lev15 = 225 hPa upper troposphere
lev16 = 250 hPa upper troposphere
lev17 = 300 hPa upper troposphere
lev18 = 350 hPa upper troposphere / mid troposphere
lev19 = 400 hPa mid troposphere
lev20 = 450 hPa mid troposphere
lev21 = 500 hPa mid troposphere
lev22 = 550 hPa mid troposphere
lev23 = 600 hPa mid troposphere
lev24 = 650 hPa mid troposphere / lower troposphere
lev25 = 700 hPa lower troposphere
lev26 = 750 hPa lower troposphere
lev27 = 775 hPa lower troposphere
lev28 = 800 hPa lower troposphere
lev29 = 825 hPa lower troposphere
lev30 = 850 hPa lower troposphere
lev31 = 875 hPa lower troposphere / boundary layer
lev32 = 900 hPa lower troposphere / boundary layer
lev33 = 925 hPa boundary layer / near-surface
lev34 = 950 hPa boundary layer / near-surface
lev35 = 975 hPa near-surface
lev36 = 1000 hPa near-surface

Rough Mapping between ERA5 levels and hPA atmospheric layers


Temporal recurrence of PC spatial patterns
PC1 2020
PC1 2021
PC2 2020
PC2 2021
PC3 2020
PC3 2021
PC4 2020
PC4 2021
PC5 2020
PC5 2021
Mean centered cosine similarity
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
250
300
350
Lag [days]